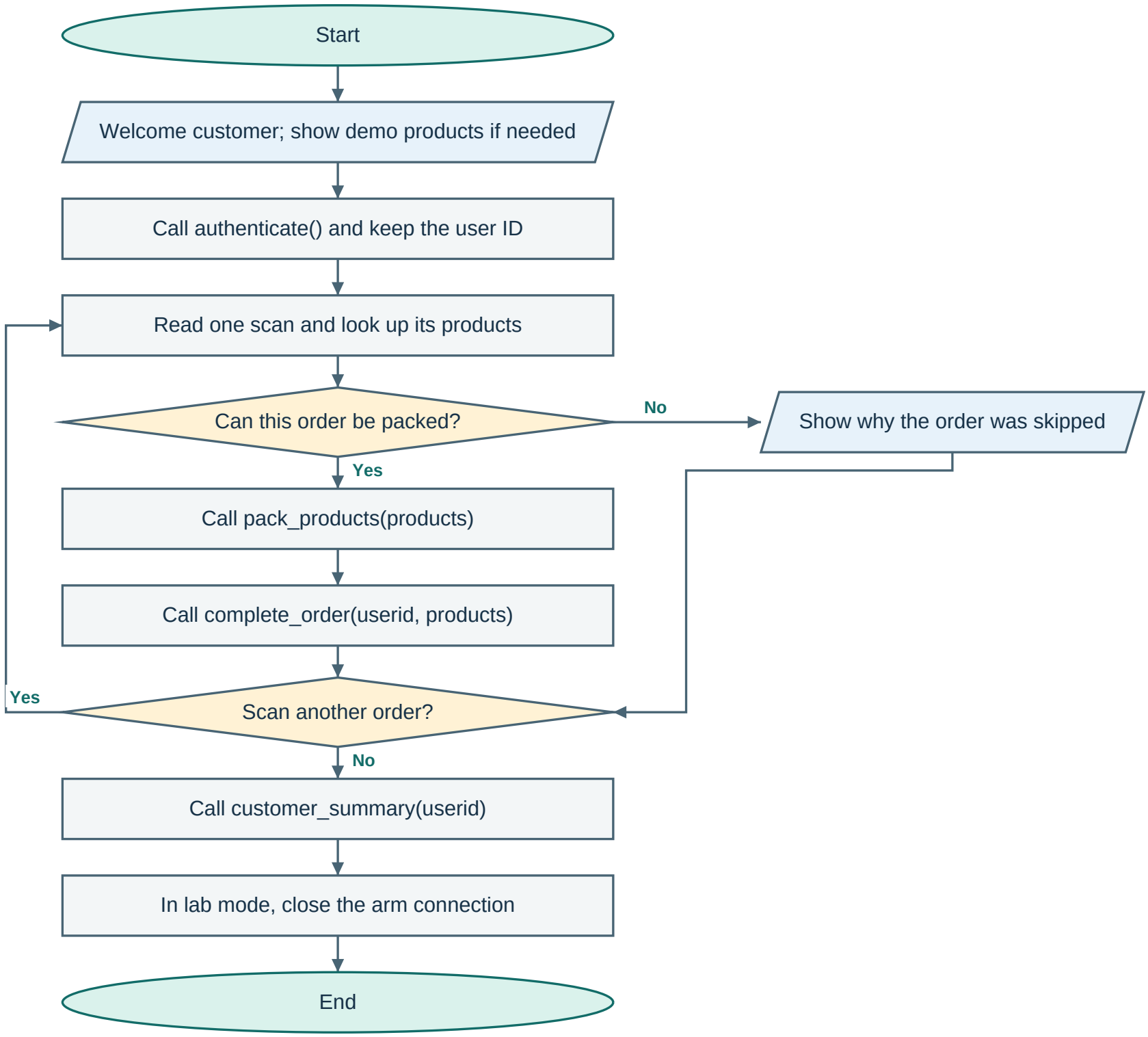
main()

Follow one customer from sign-in through their last order.



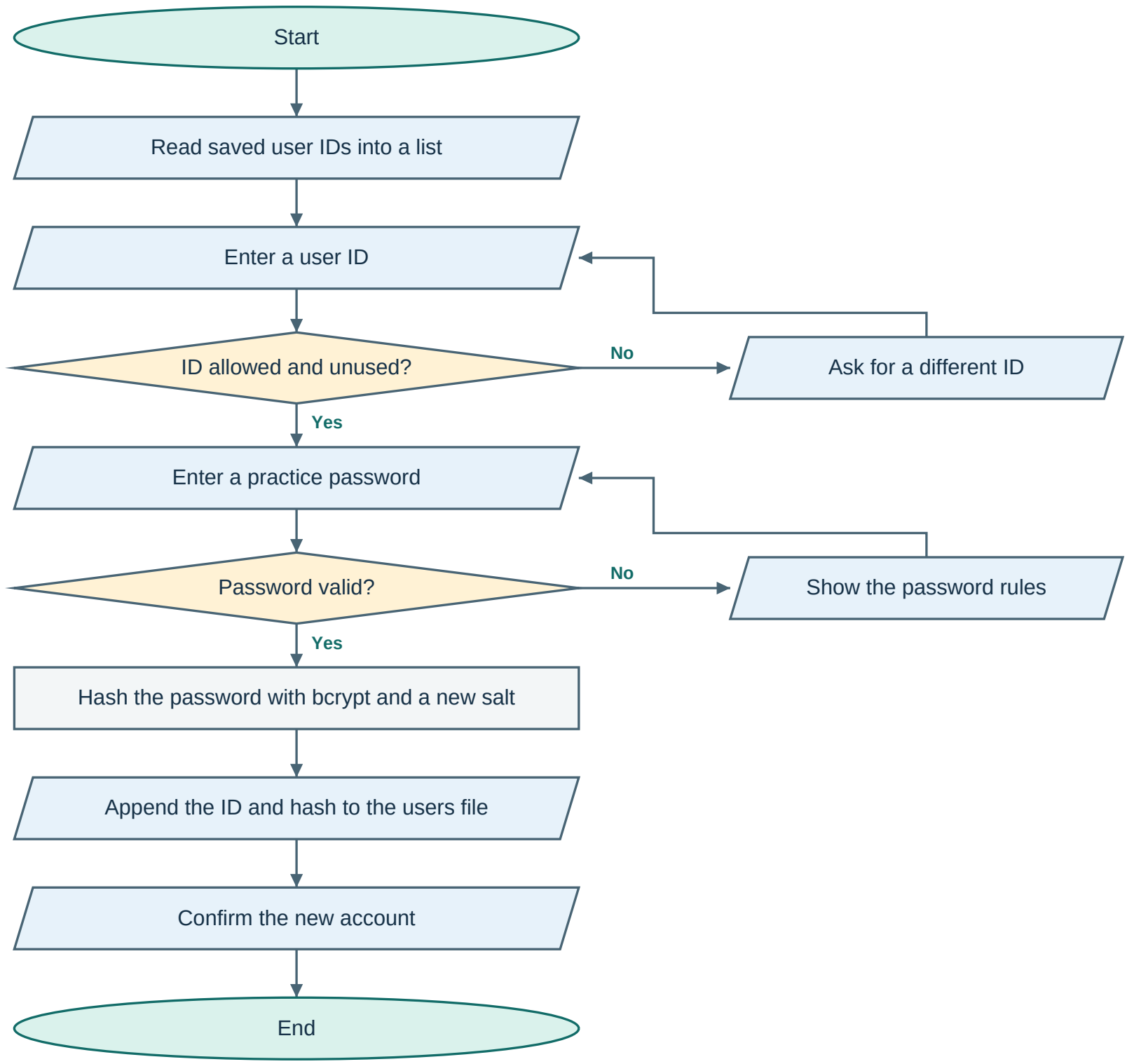
IMPLEMENTATION NOTES

can_pack / Check for matches. In the lab, also check both bin points, the closing angle and all pickups.

ask_yes_no / Repeat the prompt until the user enters y, yes, n or no.

sign_up()

Create a warehouse account using a unique ID and a valid password.



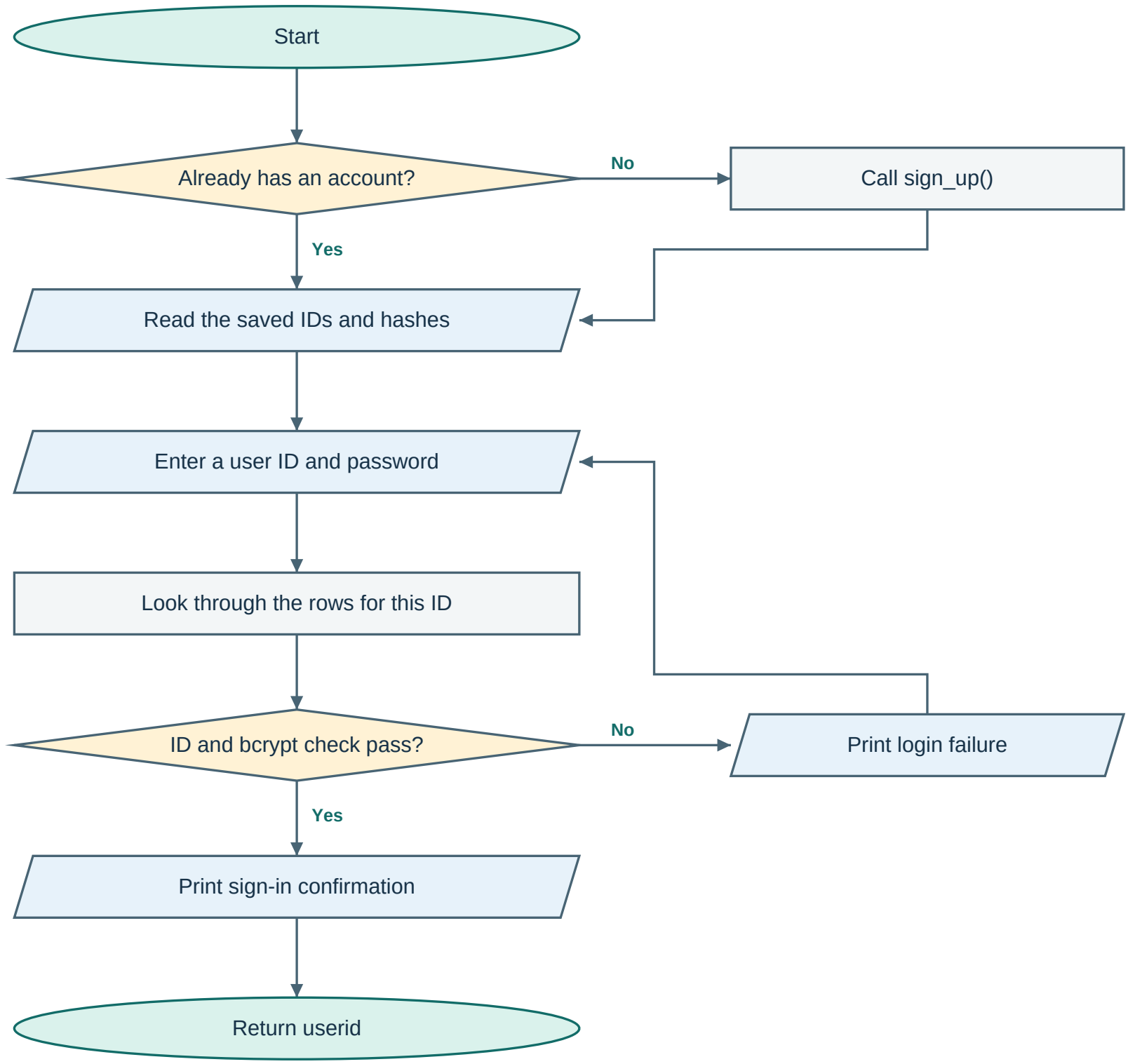
IMPLEMENTATION NOTES

ID rule / Reject empty IDs, commas and IDs already in the file.

valid_password / Test length (6+), uppercase, lowercase, digit, and one of !.@#$%^&*()_[]. Max 72 UTF-8 bytes.

`authenticate()`

Look through the stored accounts until the user signs in.



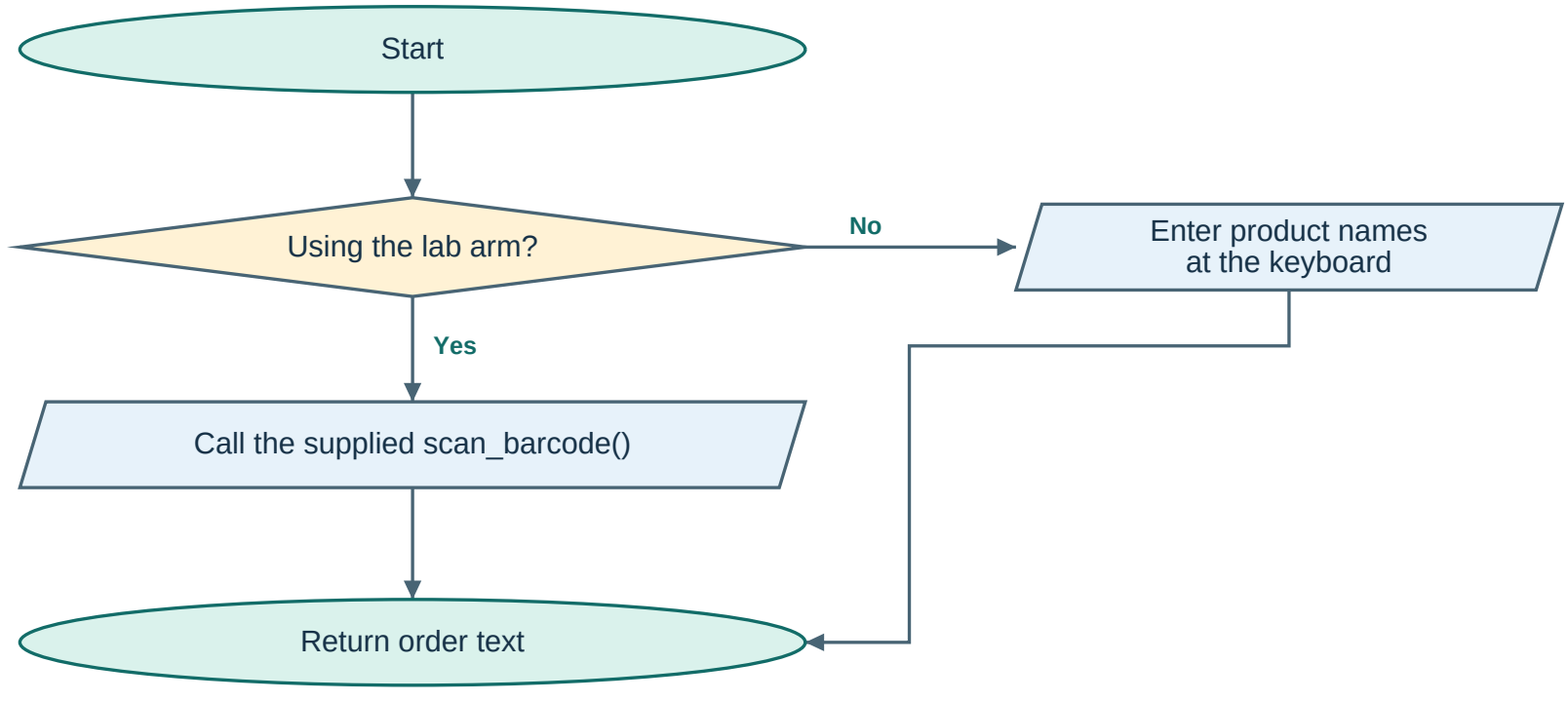
IMPLEMENTATION NOTES

Login / Check the saved hash only for a matching ID and an input within the bcrypt limit.

Returns / main() keeps the user ID for the receipt and final summary.

read_scan()

Get the comma-separated names for one order.



IMPLEMENTATION NOTES

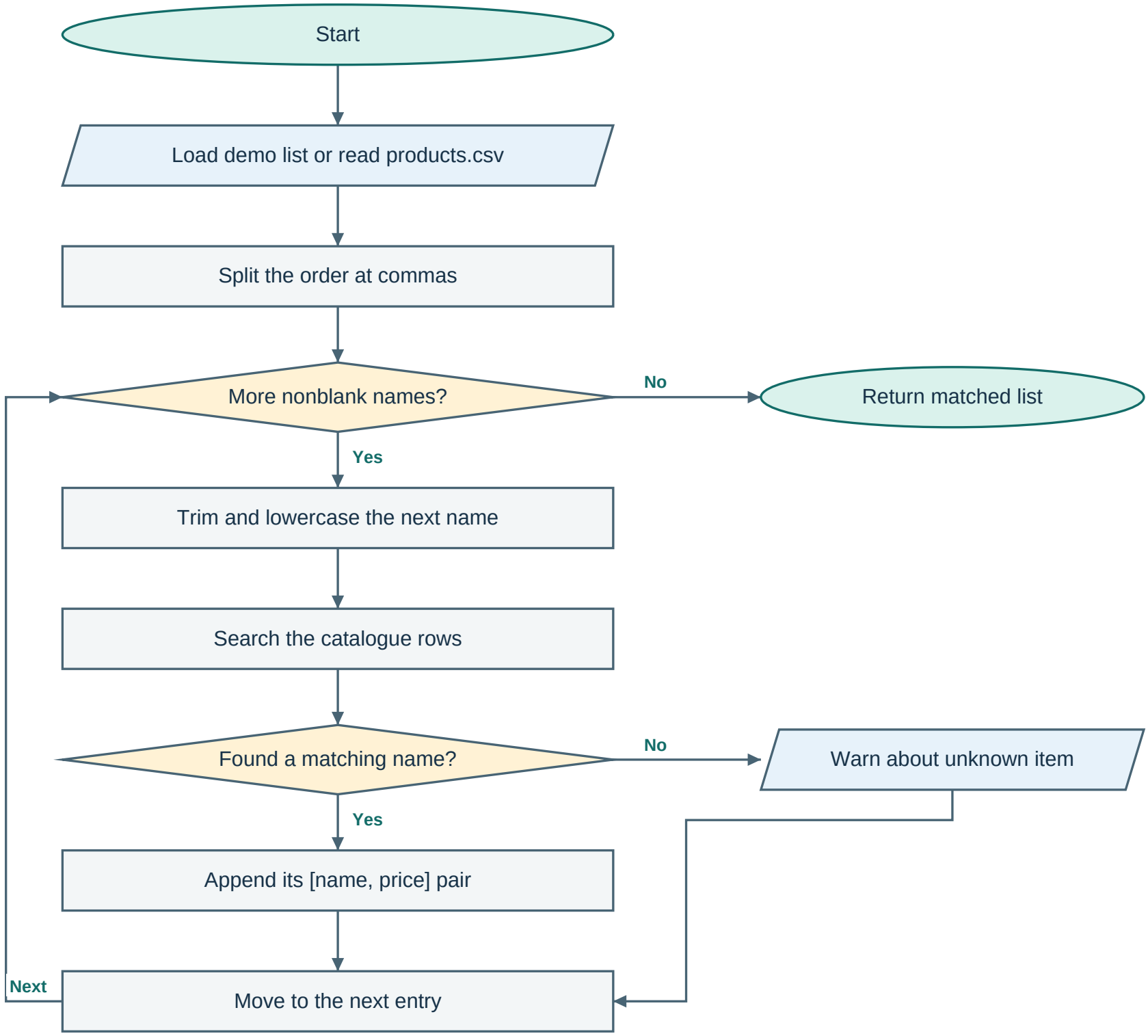
Lab / scan_barcode() is provided by the course template.

Laptop example / widget,gear,widget represents three items.

Oval Start or end / Slanted box Input or output / Rectangle Process / Diamond Decision

lookup_products(products)

Find the catalogue price for each scanned name.



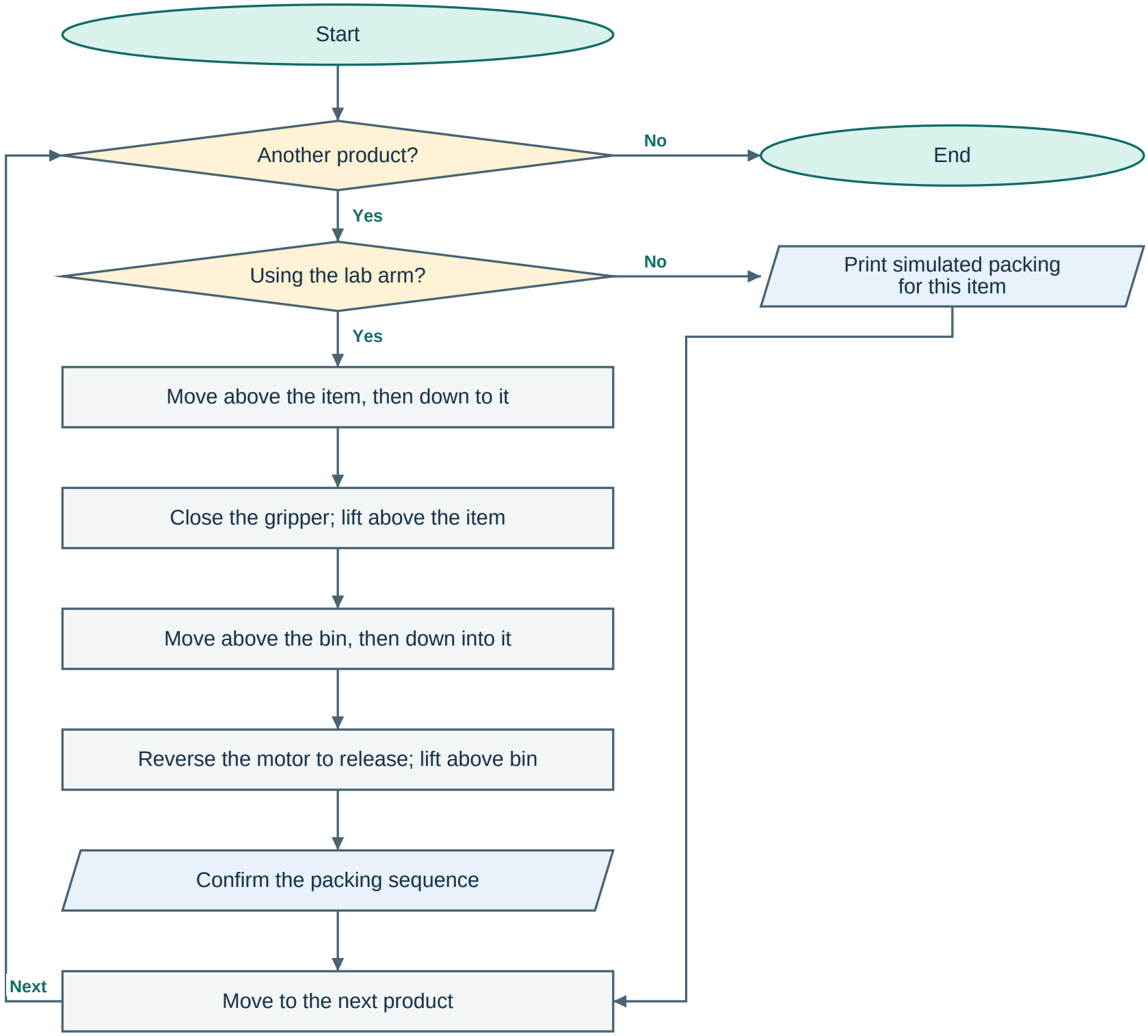
IMPLEMENTATION NOTES

Lab catalogue / Skip a price heading if present and convert the price text to a number.

Order / Keep the order of the scan and include repeated names.

pack_products(product_list)

Handle one product at a time after main() checks the order.



IMPLEMENTATION NOTES

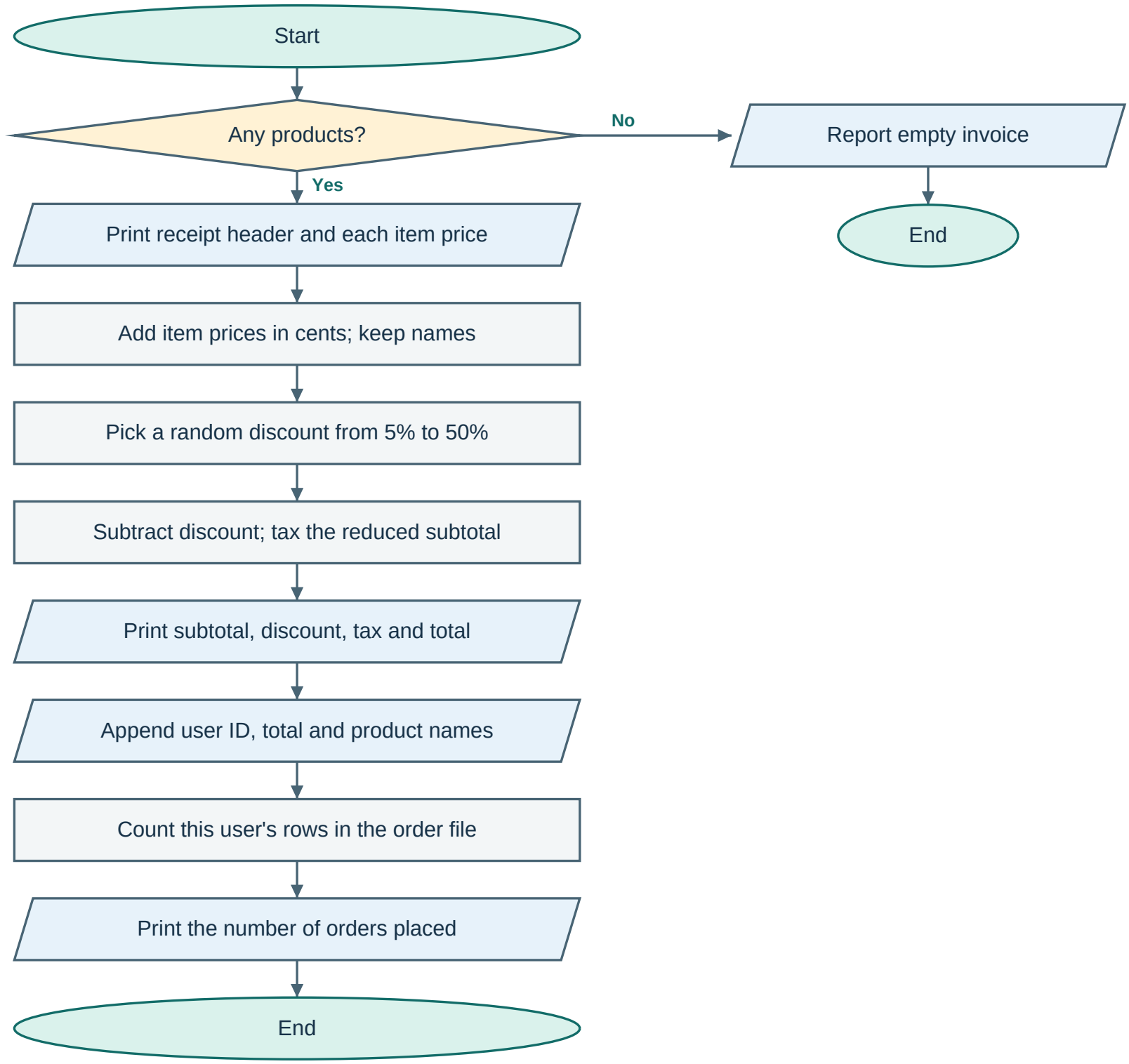
Lab setup / The team supplies measured XYZ positions and the motor angle that closes the gripper.

Timing / Wait two seconds after every arm or gripper command.

Oval Start or end / Slanted box Input or output / Rectangle Process / Diamond Decision

`complete_order(userid, product_list)`

Calculate a receipt and save one order for this customer.



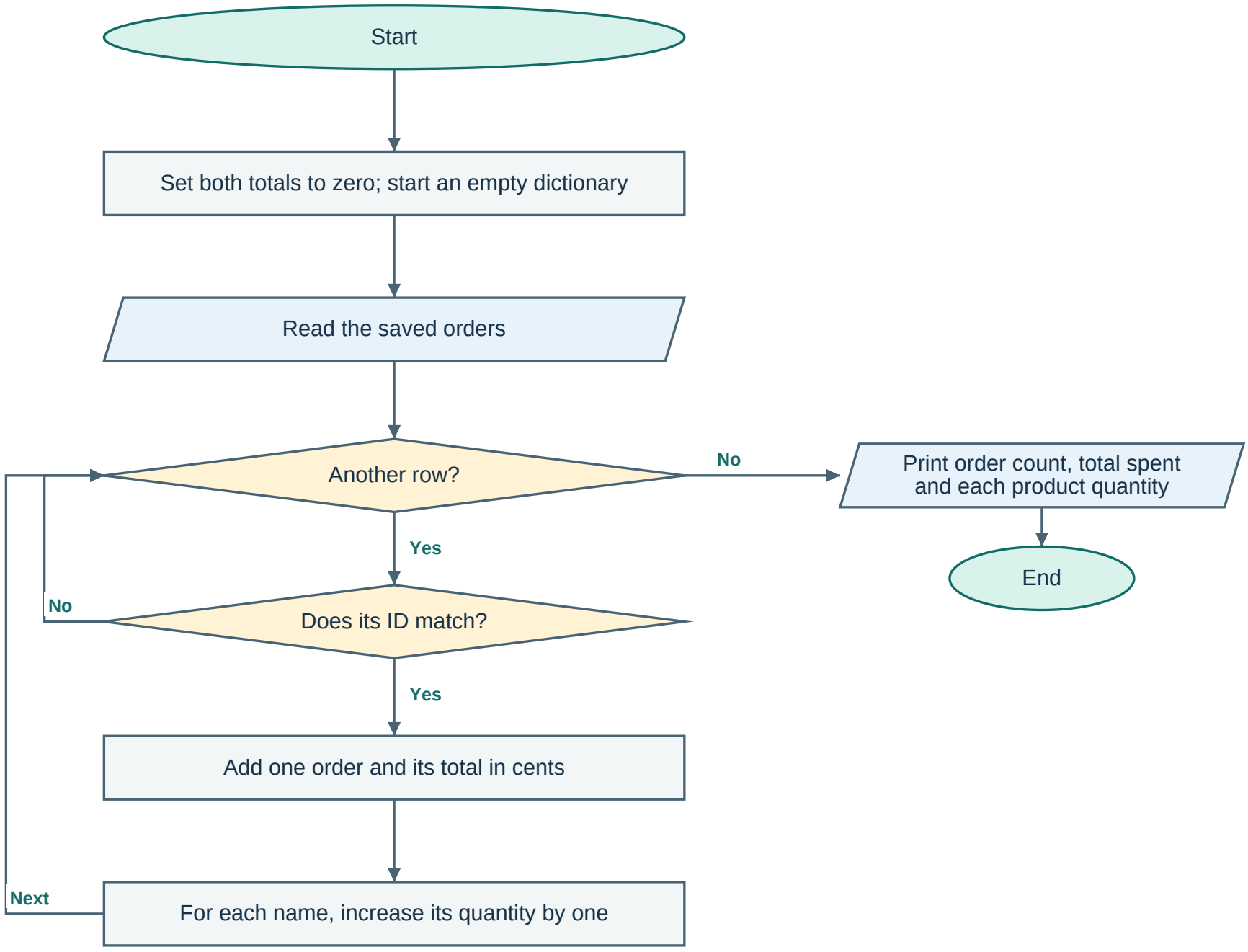
IMPLEMENTATION NOTES

Rounding / Calculate discount and tax in whole cents. The adjustable tax assumption is 13%.

Storage / Repeated products are written once per unit in the order row.

customer_summary(userid)

Accumulate spending and product quantities for one customer.



IMPLEMENTATION NOTES

Counting / Start a product at zero if it is new to the dictionary, then add one.

Empty history / Print zero orders, $0.00 spent and a no-orders message.